

SatQuery AI

Agentic Vision-Language Multi-Modal Remote Sensing Assistant

Autonomous Natural-Language Intelligence for Single, Bi-Temporal, and Cross-Modal (Optical + SAR)
Satellite Imagery

Problem Statement

Agentic Vision-Language Assistant for Multi-Modal Remote-Sensing

Theme / Category

Space Technology / AI & ML
(ISRO / SAC / MoES)

Working Prototype

LIVE & DEPLOYED

satquery-ai-psi.vercel.app

Your Team Name

SATQUERY AI



From Fragmented Satellite Tools → Intelligent Agentic Intelligence

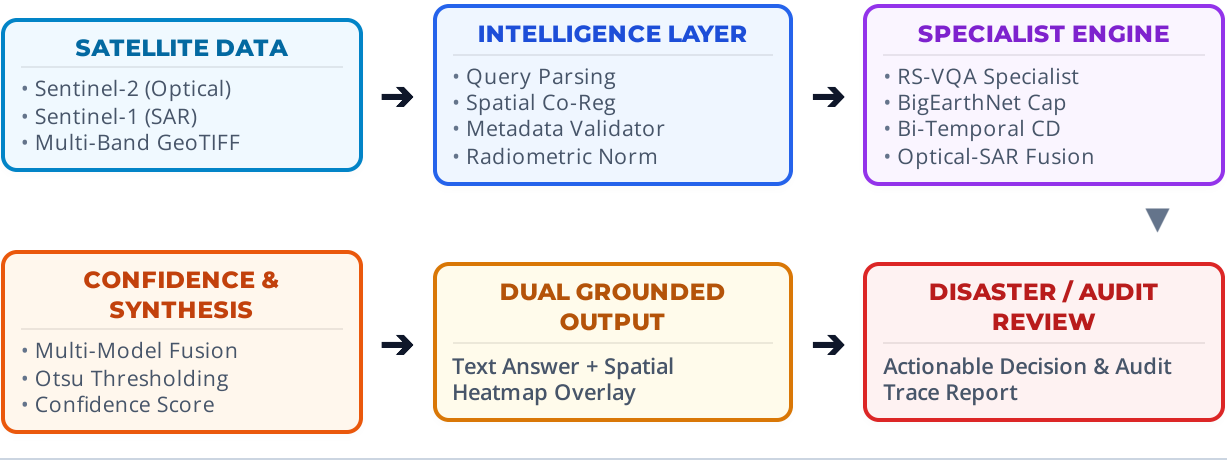
An explainable vision-language assistant that validates multi-modal imagery (Optical & SAR), dynamically chains specialized remote-sensing models, and synthesizes evidence-grounded answers.

THE PROBLEM

- Manual GIS workflows make it difficult for non-experts to query satellite imagery directly.
- Fragmented tool silos force users to know which separate model to pick (VQA, captioning, change detection).
- Weather & cloud barriers render optical imagery ineffective during monsoons and nighttime.
- Generic LLMs hallucinate land-cover features and lack verifiable spatial grounding or audit traces.

OUR SOLUTION

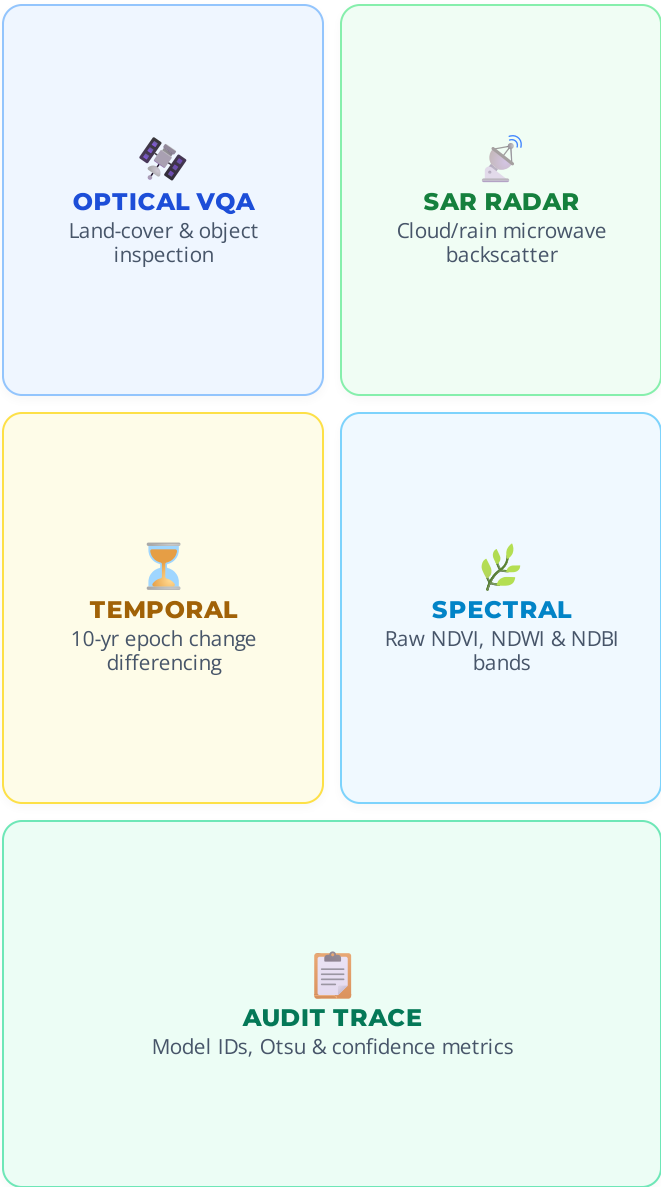
- An AI-powered agentic remote-sensing intelligence workbench.
- Ingests single-scene, multi-temporal (2016–2026), and optical + SAR pairs.
- Combines specialized vision-language models with raw spectral math.
- Detects land-cover shifts, environmental damage, and ground objects.
- Generates pixel-aligned spatial heatmaps, difference masks, and bounding boxes.
- Provides auditable execution traces and decision-support exports.



WHY DIFFERENT

- Dynamic Agentic Routing** - Automatically selects the right specialist tool based on query intent and imagery types without manual GIS setup.
- Change Understanding vs. Binary Mask** - Interprets semantic shifts (e.g. +8.5% Built-up, -14.2% Surface Water) rather than generic blur.
- All-Weather Optical-SAR Fusion** - Retains observation continuity by fusing cloud-penetrating radar when optical cloud cover exceeds 15%.
- Authentic Spectral Math** - Computes true mathematical indices (NDVI, NDWI, NDBI) on raw bands instead of generic LLM guessing.
- Verifiable Execution Trace** - Exposes an auditable card with tool IDs, thresholds, and confidence metrics for mission-critical trust.

DETECTION SIGNALS



❖ Technology Stack, Specialist AI Models & 5-Stage Implementation Methodology

Technologies to be Used

PROGRAMMING LANGUAGES, FRAMEWORKS, AI MODELS & CLOUD STACK

LANGUAGES & FRONTEND WEBGIS

Python 3.11: Core backend engine, remote-sensing raster processing & AI orchestration.
TypeScript / ES2023 & React 18 + Vite: Type-safe client architecture with sub-second HMR.
React-Leaflet & Leaflet.js: Interactive slippy map, layer toggling & split-screen swipe.
Three.js & Three-Globe: 3D WebGL orbital Earth • **HTML5 Canvas 2D:** Radiometric diff.

BACKEND, API & GEOSPATIAL CORE

FastAPI (ASGI / Python) & Uvicorn: High-throughput GeoTIFF parser with auto OpenAPI docs.
Node.js / Express (Edge Serverless): Low-latency gateway for user telemetry & routing.
GDAL, Rasterio & NumPy: Vectorized multi-band tensor math & radiometric normalization.
OpenCV, PIL, Scikit-learn, Shapely: Geometric reprojections, contours & spatial masking.

SPECIALIST AI & DEEP LEARNING MODELS

RS-VQA-Specialist: Natural language VQA on aerial/satellite textures (VRSBench / CLIP).
RS-BigEarthNet-Captioner: 19-class hierarchical CORINE land-cover taxonomy classification.
RS-BiTemporal-CD-Engine: Siamese feature differencing network (LEVIR-CD & CDVQA).
RS-CrossModal-Fusion-Core: SEN12MS & ISRO RISAT/Cartosat Optical + SAR microwave fusion.
Spectral-Band-Processor: Mathematical NDVI, NDWI & NDBI real-time index calculators.

SATELLITE FEEDS & CLOUD INFRASTRUCTURE

Satellite Feeds & APIs: Copernicus Sentinel-2 (10m Optical) & Sentinel-1 (C-band SAR), Landsat-8/9, ISRO Bhuvan archives, OpenStreetMap Nominatim.
Cloud & DevOps: Vercel Global Edge Network, Supabase Cloud Auth/DB, Docker, GitHub Actions.

Methodology & Process for Implementation

EXECUTION PIPELINE, FLOWCHART & WORKING PROTOTYPE

1. Natural Language Query & Multi-Modal Ingestion

User submits plain text prompt + AOI coordinates or uploads GeoTIFF / Optical / SAR images to multi-slots.



2. Sub-Pixel Spatial Co-Registration & Alignment

Bilinear resampling, CRS coordinate reprojection, and radiometric normalization to eliminate sun/haze bias.



3. Dynamic Specialist Routing & Execution

Agentic controller routes task to VQA, Bi-temporal change, or triggers SAR radar fusion when optical cloud cover > 15%.



4. Statistical Evidence Synthesis (Otsu Thresholding)

Adaptive Otsu thresholding over spectral differences mathematically isolates ground changes; computes confidence score.



5. Dual Grounded Output & Auditable Trace Card

Delivers scientific text summary + pixel-aligned spatial heatmap overlay + certified downloadable JSON/PDF audit report.

WORKING PROTOTYPE Live & Deployed

Multi-slot GeoTIFF uploader, 10-year satellite scrubber (2016–2026)

satquery-ai-psi.vercel.app

❖ Multi-Dimensional Feasibility Analysis, Risk Matrix & Technical Mitigation Strategies

Feasibility Analysis

TECHNICAL, OPERATIONAL & ECONOMIC DIMENSIONS

- **Technical Feasibility:** Validated by our fully operational deployed prototype ([satquery-ai-psi.vercel.app](#)) executing sub-second geocoding, TMS tile rendering, and instant AI query responses.
- **Operational Feasibility:** 100% web-based zero-install architecture; accessible on field laptops, smartphones, and emergency control rooms without local GPU hardware. Plain English interface requires zero GIS expertise.
- **Economic & Data Continuity:** Leverages free, continuous global acquisitions from Copernicus Sentinel-1/2, Landsat-8/9, and ISRO Bhuvan archives. Open-source stack eliminates proprietary GIS licensing fees (saving ₹10L+ annually per department).

Potential Challenges & Risks

IDENTIFIED TECHNICAL & OPERATIONAL BOTTLENECKS

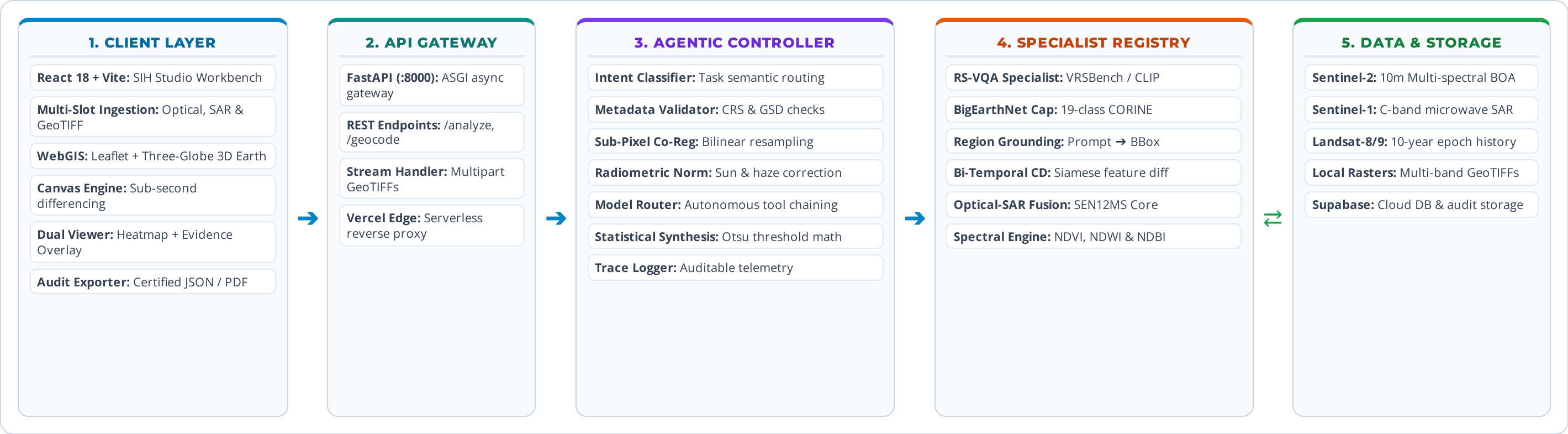
- **1. Cloud & Atmospheric Occlusion:** Heavy monsoon cloud cover, smoke, and haze obscure optical reflectance, creating blind spots during peak disasters.
- **2. Multi-Sensor Spatial Misalignment:** Ground Sampling Distance (GSD) disparities, projection distortions, and sun-angle shifts across sensors and historical epochs.
- **3. Generative Model Hallucination:** Risk of generative vision-language models producing ungrounded assertions on low-resolution or noisy satellite textures.
- **4. Heavy Raster Bandwidth Latency:** High transfer delays and memory consumption when processing full multi-gigabyte satellite GeoTIFF scenes over field networks.

Mitigation Strategies

1-TO-1 ENGINEERED TECHNICAL COUNTERMEASURES

- **1. Automated Microwave SAR Failover:** Pipeline detects >15% optical cloud cover and autonomously switches to C-band SAR radar backscatter (Sentinel-1 / RISAT) for 24/7 all-weather vision.
- **2. Sub-Pixel Affine Co-Registration:** Automated affine transformation, bilinear re-gridding, and radiometric normalization standardize all multi-temporal rasters.
- **3. Dual-Grounded Otsu Confidence Gate:** Every text answer is strictly bound to pixel-aligned spatial heatmaps and mathematical indices (NDVI/NDWI/NDBI); unverified assertions <80% are suppressed.
- **4. Pyramidal TMS & COG Streaming:** Dynamic viewport tile streaming fetches only visible sectors (~10m GSD), reducing data transfer by >90% and ensuring sub-second UI response.

❖ End-to-End Multi-Tier Agentic Architecture & Inference Pipeline



⚡ High-Throughput Streaming & Concurrency

- Pyramidal Tile Streaming:** Fetches dynamic Cloud-Optimized GeoTIFF (COG) tiles only for visible sectors, bypassing gigabyte download bottlenecks.
- Async Non-Blocking Microservice:** FastAPI asynchronous worker pool delivers sub-second inference responses to field devices.

🛡️ Zero "Black-Box" Uncertainty

- Dual Grounded Output:** Every generative textual finding is strictly corroborated by pixel-aligned spatial heatmaps and bounding-box evidence.
- Deterministic Mathematical Grounding:** Otsu statistical thresholding and raw band spectral ratios prevent hallucination.

📡 All-Weather Multi-Modal Failover

- Automated Radar Switching:** Pipeline triggers C-band SAR backscatter when cloud screening detects >15% cloud occlusion.
- Continuous 24/7 Operations:** Retains mission-critical observations through monsoons, cloud cover, and total darkness.

Potential Impact on Target Audience

- **Disaster Management Agencies (NDRF / SDMA's):** Instantaneous post-disaster flood inundation extent and structural damage delineation in minutes instead of days.
- **Urban Planning & Municipal Corporations:** Automated detection of unauthorized construction, peripheral urban sprawl (+14.2% detected in benchmarks), and encroachment on green buffers across 10-year timelines.
- **Agriculture & Irrigation Departments:** Automated crop phenology tracking (Kharif/Rabi vegetative vigor), drought stress detection, and reservoir water retreat monitoring.
- **Defense & Strategic Border Surveillance:** All-weather day-and-night change monitoring using cross-modal Optical + SAR backscatter signatures.
- **Environmental Authorities:** Rapid monitoring of illegal deforestation, wetland shrinkage (e.g. Ramsar Bhoj Wetland), and urban heat islands.

Societal, Economic & Environmental Benefits

- **Social Benefit:**
 - Democratizes space technology: field officers, district collectors, and researchers can query Earth observation data in plain English without technical GIS training.
 - Enables rapid, transparent governance and disaster relief deployment.
- **Economic Benefit:**
 - Eliminates recurring proprietary GIS software licensing fees (saving lakhs per government department).
 - Reduces satellite imagery interpretation turnaround time by over 80%.
- **Environmental Benefit:**
 - Continuous monitoring of ecologically sensitive zones, water conservation bodies, and forest reserves.
 - Actionable spatial intelligence to support Sustainable Development Goals (SDGs 11, 13, 15).

Scientific Benchmarks & Literature

- **RSVQA Benchmark:** Lobry et al., "*RSVQA: Visual Question Answering for Remote Sensing Data*", IEEE Transactions on Geoscience and Remote Sensing (TGRS).
- **VRSBench Dataset:** Chen et al., "*VRSBench: A Versatile Vision-Language Benchmark for Remote Sensing Image Understanding*", IEEE/CVF CVPR 2024.
- **LEVIR-CD & CDVQA:** Chen & Shi, "*A Spatial-Temporal Attention-Based Method and Dataset for Remote Sensing Image Change Detection*", Remote Sensing.
- **SEN12MS Dataset:** Schmitt et al., "*SEN12MS — A Curated Dataset of Georeferenced Multi-Spectral Sentinel-1/2 Imagery for Deep Learning and Data Fusion*", ISPRS Annals.
- **BigEarthNet-S2:** Sumbul et al., "*BigEarthNet: A Large-Scale Benchmark Archive for Remote Sensing Image Understanding*", IEEE IGARSS.

Space Agencies & Data Standards

- **ISRO / SAC / NRSC:** Technical standards for Cartosat High-Resolution Optical and RISAT-1A C-band Synthetic Aperture Radar (SAR) processing protocols.
- **ISRO Bhuvan Geoportal:** Open geospatial data standards and national thematic map services.
- **ESA Copernicus Open Access Hub:** Sentinel-2 MultiSpectral Instrument (MSI) Level-2A Bottom-of-Atmosphere (BOA) reflectance specifications.
- **SpatioTemporal Asset Catalog (STAC):** Standardized metadata specifications for multi-epoch satellite imagery indexing.
- **Open Geospatial Consortium (OGC):** Compliance with WMS, TMS, and Cloud-Optimized GeoTIFF (COG) standards.

Live Prototype & Open Source Links

- **Live Deployed Prototype Application:**
<https://satquery-ai-psi.vercel.app/>
100% OPERATIONAL ON VERCEL EDGE
- **Official GitHub Source Repository:**
<https://github.com/sourabh160705/Satquery-AI>
OPEN-SOURCE CODEBASE
- **API Endpoint Documentation:**
Interactive Swagger / OpenAPI docs for SIH evaluators.
- **Full Audit Trail Export:**
Standardized JSON & GeoJSON execution proofs downloadable directly from the SIH Studio console.